Cheedella Vamsi Kishore

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Professional Summary

AI Engineer with experience in developing and deploying LLM APIs, vector databases, and Retrieval-Augmented Generation (RAG) applications in production. Proficient in Machine Learning (ML), AI Agents, Natural Language Processing (NLP), and Computer Vision (CV), with a solid foundation in problem-solving, optimization, and data structures and algorithms. Holds a Codeforces rating of 1000+. Skilled in building proof-of-concept (PoC) solutions and transitioning them into scalable production systems. Adept at troubleshooting production-level RAG application issues and optimizing performance. Strong collaborator with cross-functional teams to deliver data-driven, impactful applications. Expertise in Python, TensorFlow, PyTorch, and Azure.

Experience

Associate Consultant Sept 2022- Present
Capgemini *Hyderbad, Telangana*

AI-Driven Agile Workflow Automation

- Our current product assumes to increase efficiency of scrum master by around **45% in large teams and 30% in small teams**
- Results-driven Senior Python Developer with expertise in AI-driven automation, (RAG) Crew AI, and AI security using Flask, ChromaDB, Azure GPT API and Azure Devops.
- **Doc Intelligence**
  * **InsightFlow:** Implemented Iterative summarize for structured and unstructured document processing handling 30,000+ words per document using semantic chunking and a fixed-length sliding window (70k tokens, 20% overlap) for context retention.
  * **Ask Doc :** Given an input prompt, we apply semantic chunking (for structured docs) or fixed-length chunking with overlap (for unstructured text) to retain context. Using vector search, we retrieve the most relevant chunks and combine them with the overall document summary. Finally, the GPT API generates an accurate response based on the refined context.
- **Deduplication**
  * Implemented a vector database–driven system to identify and link similar test cases in Azure DevOps (ADO), reducing redundancy and improving test coverage.
  * Leveraged semantic similarity using embeddings of test case titles and descriptions to automatically detect and associate related cases. This enhanced traceability and streamlined test management within ADO.
- **Root Cause Analysis for bugs**
  * RAG Based approach to find the related userstory and testcase and analyze it
  * Implemented role-based LLM prompting (Developer, QA, PM) to identify failure points, improving bug resolution efficiency.
- **Test case Prioritization** - find Complexity, Dependency, and Criticality scores for testcases
  * Analyzed 10,000+ test cases from Azure DevOps using Azure GPT API to assign prioritization scores.
  * Leveraged asynchronous function calls for parallel processing, accelerating test case scoring by 5x and streamlining test planning for faster decision-making.
- **Response Shield AI**
  * **Multi turn Testing:** Validates the chatbot’s ability to maintain coherent and responsible interactions over each turn, ensuring consistency and ethical compliance across a conversation. Measures answer relevancy, bias, toxicity and hallucinations in conversations.
- **Agile Intelligence**
  * GPT-driven system to automate Agile artifacts with a structured, scalable approach.
  * Implemented structured and few-shot prompting for consistency, role-based prompting for stakeholder alignment, and Chain of Thought (CoT) prompting for test case generation.

Life Insurance Prediction - production-ready model to predict life insurance premiums using LGBM

- Engineered features using correlation analysis and domain expertise; **reduced feature set from 18 to 14 by creating task-specific features while preserving predictive signal**. Used SHAP values to validate model interpretability for stakeholders.
- Benchmarked boosting algorithms (XGBoost, CatBoost, GradientBoosting) and **chose LightGBM for its high accuracy and 3–4× faster training speed**; optimized hyperparameters with Optuna to reduce RMSLE during cross-validation.

Technical Skills

Languages	:	C, CPP, Python, SQL
Cloud services	:	Azure Blob Storage, App service, Azure SQL, Azure OpenAI
Frameworks	:	TensorFlow, Keras, Pytorch, HuggingFace
Libraries	:	Scikit-Learn, numpy, cv2, xgboost, Numpy, Pandas,Langchain, Langgraph, CrewAI
Dev Tools	:	Git, Docker, Visual Studio Code
Other Skills	:	Linux, Data Structures

Education

National Institute of Technology, Calicut Calicut, Kerala
B.Tech Electrical and Electronics Engineering, CGPA: 8.17/10 *Aug 2018 – May 2022*

Projects

- **Invoice Analyzer Bot using gemma**
  - Developed a chatbot capable of interacting with structured and unstructured invoice and purchase bill documents.
  - Utilized Gemma 12B multimodal model to extract key-value pairs and tabular data from invoices and store them in JSON format.
  - Implemented a ReAct-style prompting strategy integrated with tools such as JsonQuery, Tavily, and Calculator to ensure accurate and context-aware responses strictly grounded in the provided invoice.
  - Designed a dynamic tool selection mechanism: the bot analyzes user queries, invokes the most relevant tool based on context, and iteratively selects subsequent tools until a conclusive answer is derived or the goal is met.
- **LangGraph CodeTutor: Baby AGI-Inspired Educational Agent**
  - A LLM-powered code generation system with planning and evaluation capabilities based on the ReAct framework using LangGraph and LangChain, to iteratively plan, generate, and evaluate Python code with a focus on beginner education.
  - Engineered a stateful graph with conditional edges to manage transitions between planning, code writing, evaluation, and replanning, ensuring controlled agent behavior.
  - Built an automated evaluation framework that executes code and delivers instructor-like feedback for guided, step-by-step learning and improvement. Created custom ReAct parsers to preserve execution flow when LLM outputs deviate from expected formats.
  - Generated detailed execution logs showcasing programming methodology and decision-making process for beginners. Built system that maintains code continuity across iterations, demonstrating how to incrementally improve solutions rather than rewriting from scratch.
- **Underwater image enhancement**
  - Developed an autoencoder-based image enhancement model to remove underwater blue tint, **improving visibility by 45% for better localization and object detection.**
  - Applied **advanced image processing techniques for object extraction** and designed a control system to support autonomous navigation in underwater environments.

Certifications

- **Coursera:** Deep learning Specialization, Self-driving cars Specialization.
- **Electives:** AI and fuzzy logic, Digital signal processing, Control Systems